

Mathematical Modeling of Rocket Landing System

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1. Introduction

Reusable rockets are advanced aerospace systems designed to perform controlled vertical landing after flight. These systems use thrust vector control and feedback-based stabilization to achieve precision landing.

To analyze the motion and stability of the rocket, a mathematical model is required that relates **forces, torques, and resulting motion**.

In this work, the equations of motion are derived using the **Lagrangian method**, which is widely used in dynamic system modeling and control.

2. System Description

The rocket is modeled as a **rigid body in planar motion**.

Degrees of Freedom (DOF):

- Horizontal motion $\rightarrow x$
- Vertical motion $\rightarrow z$
- Rotation $\rightarrow \theta$

Inputs to the system:

- Thrust T
 - Gimbal angle δ
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3. Modeling Assumptions

To simplify the modeling process:

1. Rocket is treated as a rigid body
2. Motion is planar (yaw ignored)

3. Small angle approximation used
 4. Thrust acts along body axis
 5. Drag is proportional to velocity
 6. Mass is constant
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4. Coordinate System and Generalized Coordinates

A coordinate system is defined:

- x -axis $\rightarrow$ horizontal
- z -axis $\rightarrow$ vertical

Generalized coordinates:

$$q = [x, z, \theta]$$

These variables completely define the system configuration.

5. Choice of Dynamic Modeling Method

The **Lagrangian formulation** is used because:

- It simplifies multi-DOF systems
 - Avoids direct calculation of constraint forces
 - Provides a systematic energy-based approach
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6. Kinematic Description

Position of rocket center of mass:

$$x = x(t), z = z(t)$$

Orientation:

$$\theta = \theta(t)$$

Velocities:

$$\dot{x}, \dot{z}, \dot{\theta}$$

Accelerations:

$$\ddot{x}, \ddot{z}, \ddot{\theta}$$

7. Forces Acting on the System

1. Thrust Force:

$$T_x = T \sin (\theta + \delta)$$
$$T_z = T \cos (\theta + \delta)$$

2. Gravity:

$$F_g = mg$$

3. Drag Forces:

$$F_x = -c\dot{x}, F_z = -c\dot{z}$$

4. Torque due to thrust:

$$\tau = lT \sin (\delta)$$

8. Kinetic Energy

$$T = \frac{1}{2} m(\dot{x}^2 + \dot{z}^2) + \frac{1}{2} I \dot{\theta}^2$$

9. Potential Energy

$$V = mgz$$

10. Lagrangian Formulation

$$L = T - V$$

$$L = \frac{1}{2}m(\dot{x}^2 + \dot{z}^2) + \frac{1}{2}I\dot{\theta}^2 - mgz$$

11. Equations of Motion

Using:

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right) - \frac{\partial L}{\partial q_i} = Q_i$$

We obtain:

$$\begin{aligned} m\ddot{x} &= T\sin(\theta + \delta) - c\dot{x} \\ m\ddot{z} &= T\cos(\theta + \delta) - mg - c\dot{z} \\ I\ddot{\theta} &= lT\sin(\delta) \end{aligned}$$

12. Final Dynamic Model

The system can be written as:

$$\begin{cases} \ddot{x} = \frac{T\sin(\theta + \delta) - c\dot{x}}{m} \\ \ddot{z} = \frac{T\cos(\theta + \delta) - mg - c\dot{z}}{m} \\ \ddot{\theta} = \frac{lT\sin(\delta)}{I} \end{cases}$$

13. State-Space Representation

Define state vector:

$$\begin{aligned} x &= [x, \dot{x}, z, \dot{z}, \theta, \dot{\theta}] \\ \dot{x} &= \begin{bmatrix} \dot{x} \\ \frac{T\sin(\theta + \delta) - c\dot{x}}{m} \\ \dot{z} \\ \frac{T\cos(\theta + \delta) - mg - c\dot{z}}{m} \\ \dot{\theta} \\ \frac{lT\sin(\delta)}{I} \end{bmatrix} \end{aligned}$$

14. Unit Consistency Check

Quantity	Unit
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Position	m
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Velocity	m/s
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Acceleration	m/s ²
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Force	N
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Torque	N·m
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Mass	kg
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All equations are dimensionally consistent.

15. Physical Meaning of Terms

- $T \sin (\theta + \delta) \rightarrow$ horizontal thrust
 - $T \cos (\theta + \delta) \rightarrow$ vertical thrust
 - $c\dot{x}, c\dot{z} \rightarrow$ drag forces
 - $mg \rightarrow$ gravity
 - $lT \sin (\delta) \rightarrow$ control torque
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16. Conclusion

A complete dynamic model of the rocket landing system was derived using the Lagrangian method.

These equations describe how **thrust and control inputs affect motion**, forming the basis for **simulation, control design, and landing stability analysis**.